

$$\alpha = \delta_x$$

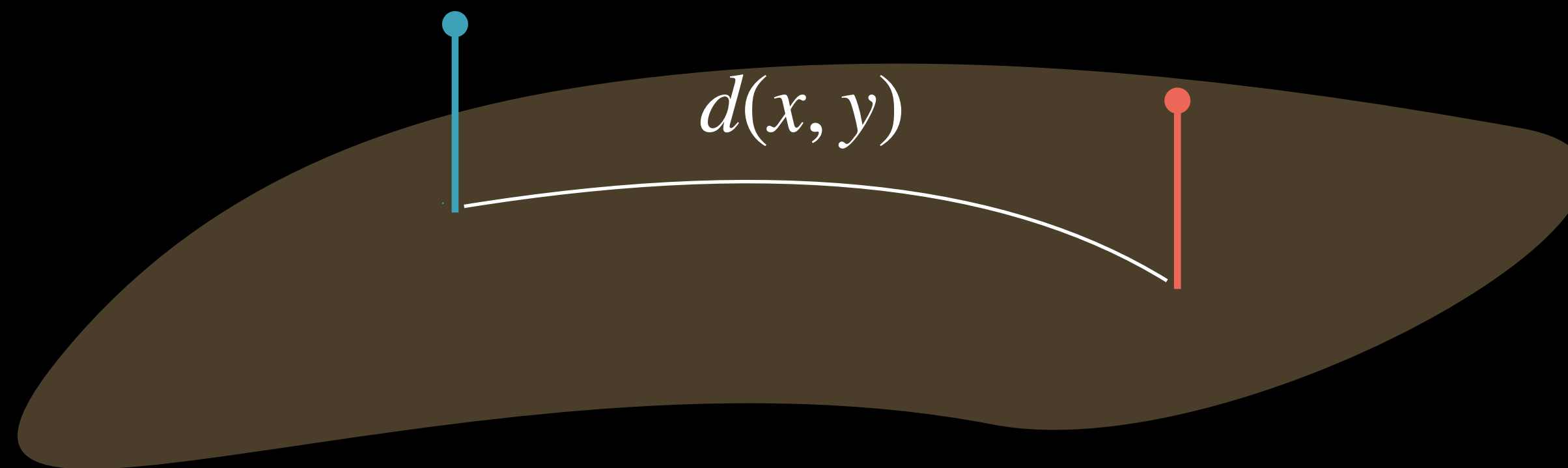
$$\beta = \delta_y$$



Example: Diracs

$$\alpha = \delta_x$$

$$\beta = \delta_y$$



$$W_p(\alpha, \beta) = \left(\inf_{\pi \in \Pi(\alpha, \beta)} \int_X d(x, y)^p \, d\pi(x, y) \right)^{1/p} \rightarrow W_p(\alpha, \beta) = d(x, y)$$

Good sanity check! Wasserstein reduces exactly to the ground metric, is “consistent”

W_2 between 1D Gaussians

$$\alpha = \mathcal{N}(\mu_\alpha, \sigma_\alpha^2) \quad \beta = \mathcal{N}(\mu_\beta, \sigma_\beta^2)$$

